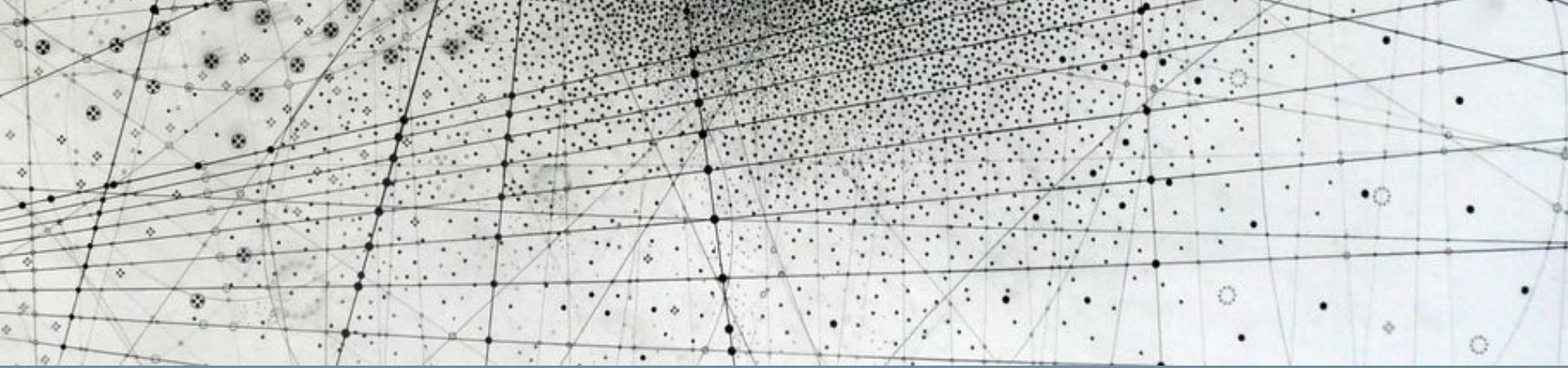


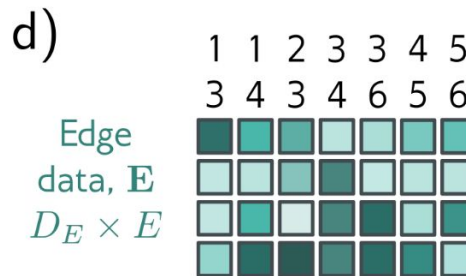
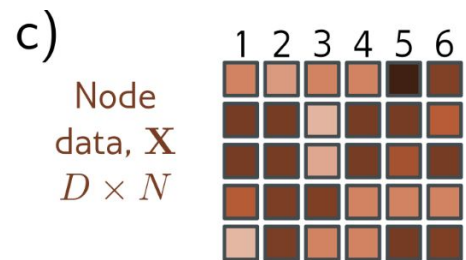
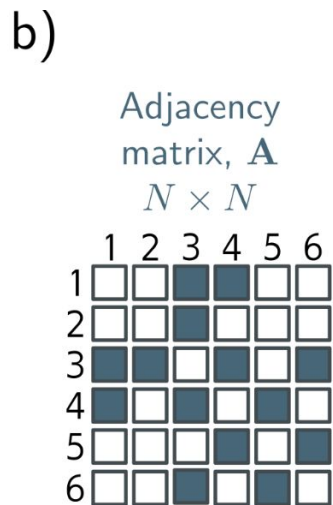
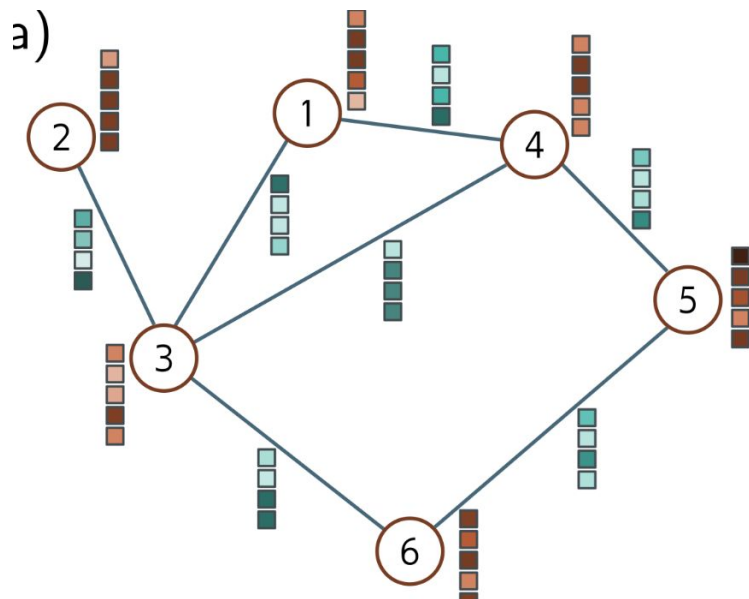


Research Methods in Machine Learning & Physics

Quick recap

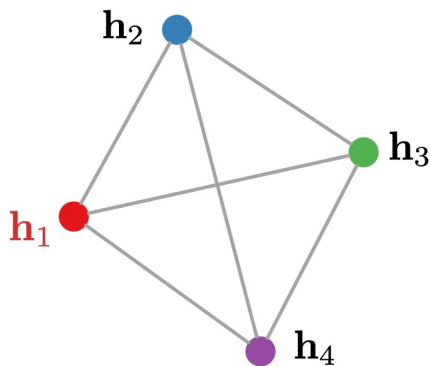
Quick Recap: Graph Neural Networks

- “Graph data” = collections of nodes (objects) and edges (relationships)
- We represent these in neural networks via a set of matrices:
 - **Adjacency matrix:** specifies the edges connecting nodes
 - **Node feature matrix:** (# of nodes) x (# of node features)
 - **Edge feature matrix:** (# of edges) x (# of edge features)

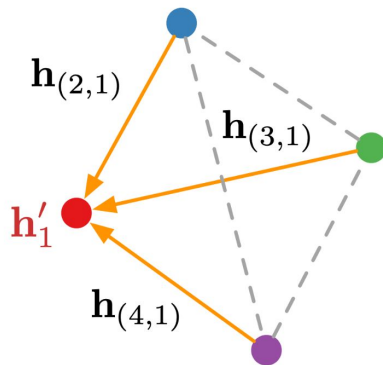


Quick Recap: Graph Neural Networks

- **GNNs can solve many tasks, including:**
 - Graph-level classification or regression (graph $\rightarrow$ label or number)
 - Node-level classification or regression (node $\rightarrow$ label or number)
 - Edge-level classification or regression (e.g. edge prediction)
- GNNs apply graph convolutions via **message passing** steps that make updates to the graph



(a) Node representations.



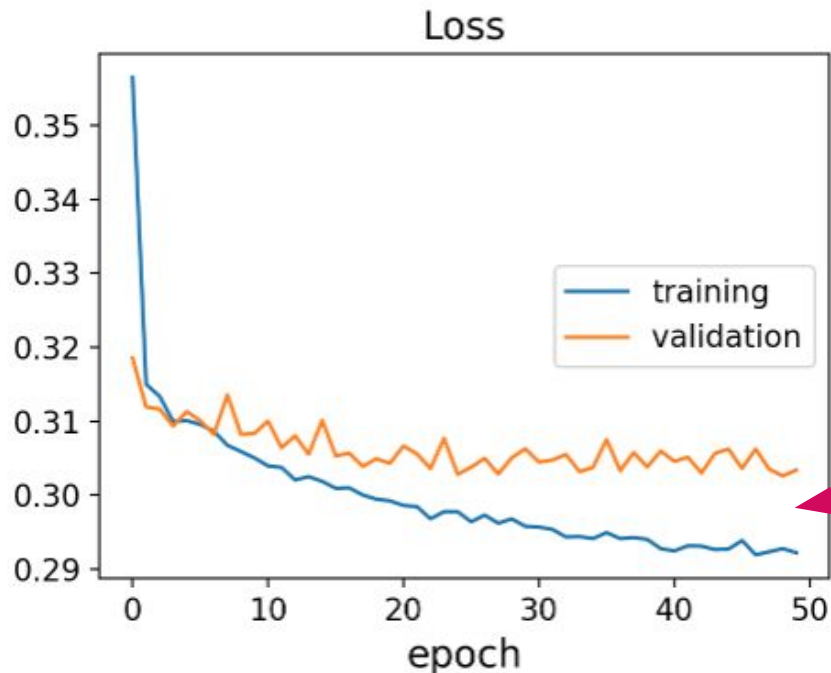
(b) Message passing step.

$$\mathbf{h}_{(i,j)} = f_{\text{edge}}(\mathbf{h}_i, \mathbf{h}_j, \mathbf{x}_{(i,j)}),$$
$$\mathbf{h}'_i = f_{\text{node}}(\mathbf{h}_i, \sum_{j \in \mathcal{N}_i} \mathbf{h}_{(j,i)}, \mathbf{x}_i)$$

Regularization

Regularization helps reduce overtraining.

- **Overtraining** is a problem frequently encountered when your model is too powerful for the number of training data samples you have available to use.

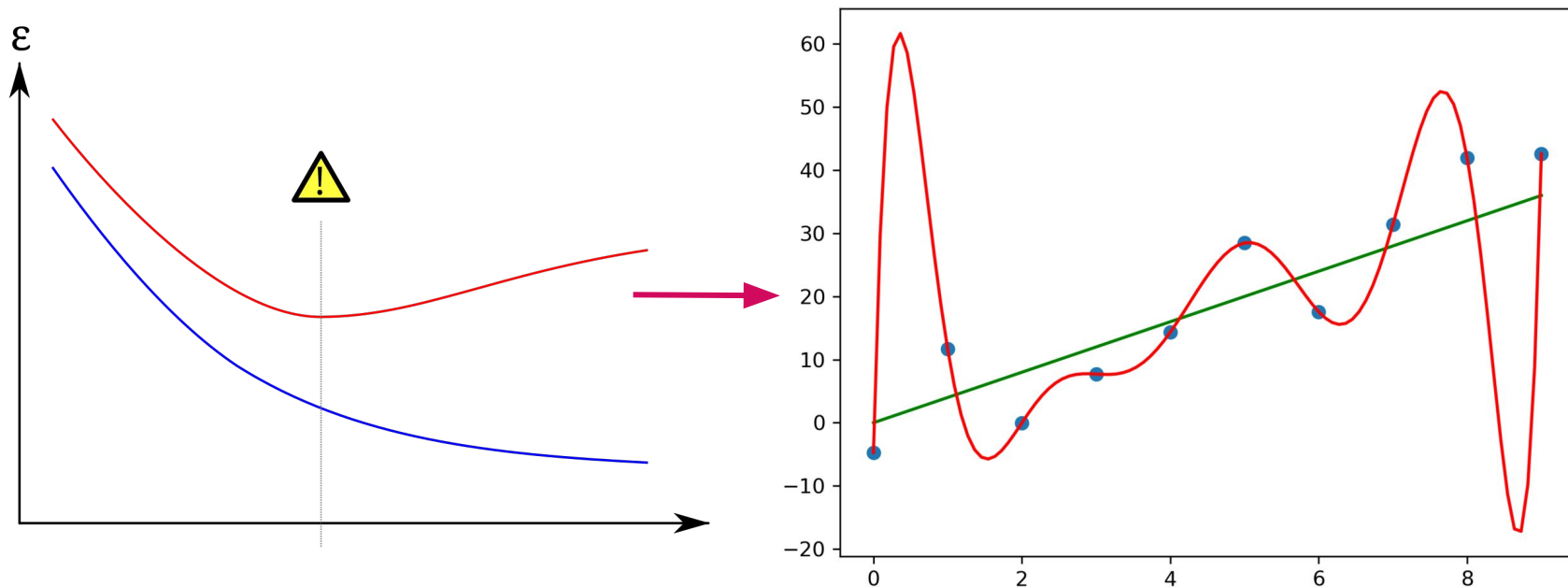


Overtraining:

- Training loss keeps going down
- Validation loss stays constant
- Gap between them grows
- Your model is memorizing 🤔

Overtraining is an indication of overfitting:

- When your network is overfitting, it's learning an overly-complicated function and won't generalize well.



Regularization = improving generalization!

Explicit regularization: adding a penalty to the loss

A very common example is **L2 regularization**:

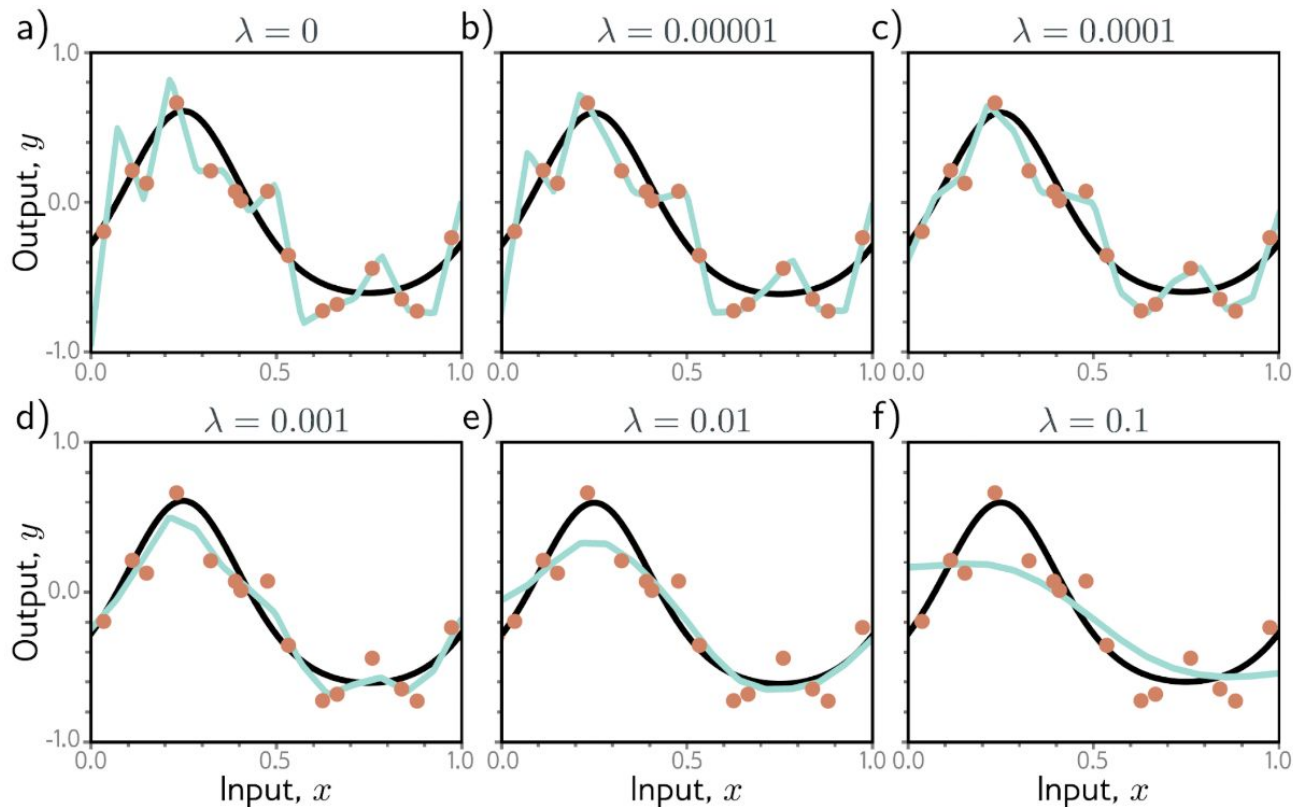
- Also called L2 norm, Tikhonov regularization, ridge regression, weight decay...

$$\hat{\phi} = \operatorname{argmin}_{\phi} \left[L[\phi, \{\mathbf{x}_i, \mathbf{y}_i\}] + \lambda \sum_j \phi_j^2 \right]$$

Coefficient

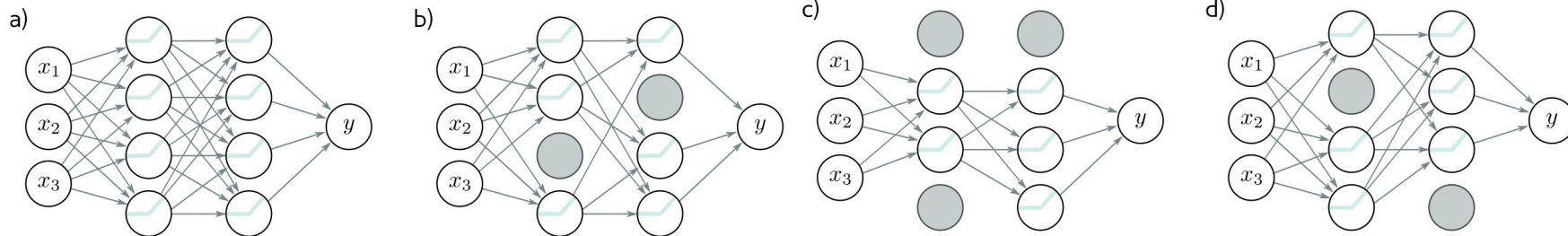
Penalty to the loss
term if the weights
get too large

Explicit regularization: adding a penalty to the loss

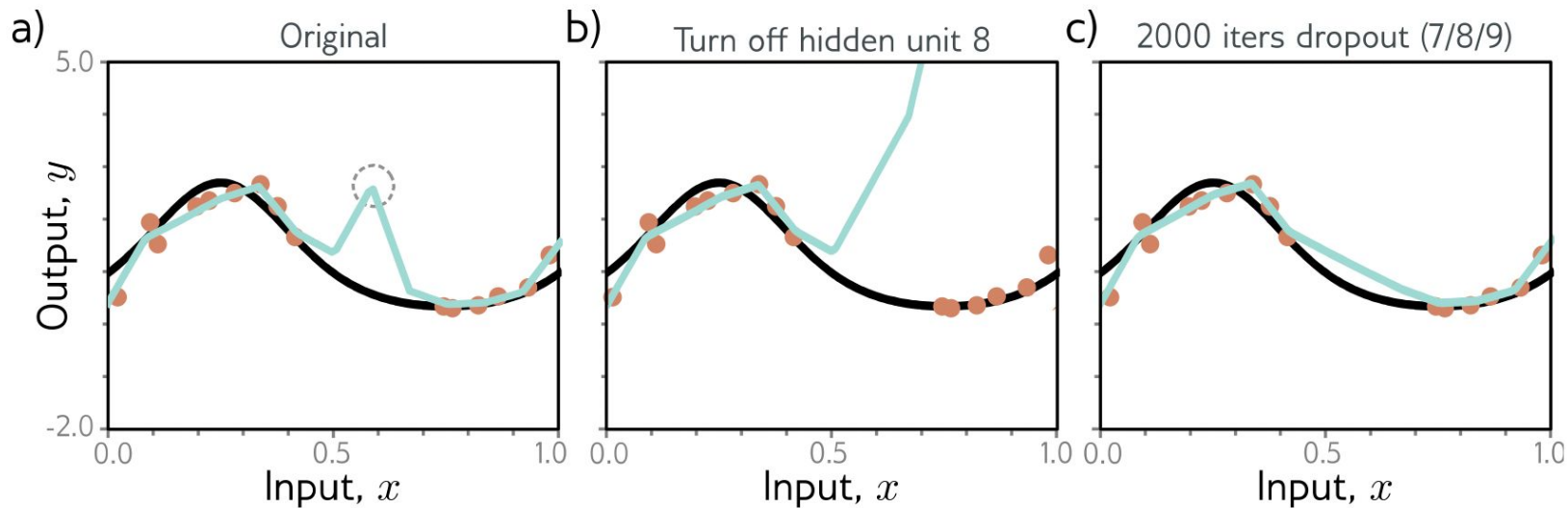


Explicit regularization: dropout layers

Randomly kill (i.e. zero-out) some nodes in each layer according to your specified "dropout probability" (a hyperparameter choice):



Explicit regularization: dropout layers



Can help make your model more robust to outliers/noise.

Implicit regularization:

Naturally-regularizing effects emerging from your model/training paradigm

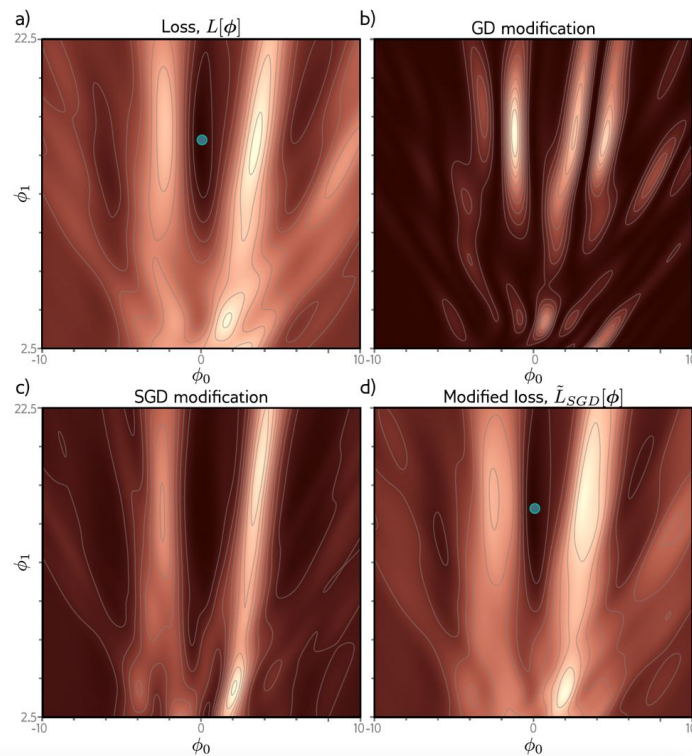
Gradient descent (GD) and Stochastic Gradient Descent (SGD) are **not neutral** – they have preferences for certain solutions through the loss landscape.

For instance, GD with a finite step size will end up slightly avoiding areas with steep gradients:

$$\tilde{L}_{GD}[\phi] = L[\phi] + \frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2$$

Similarly, SGD ends up preferring that all batches have roughly the same gradients:

$$\begin{aligned} \tilde{L}_{SGD}[\phi] &= \tilde{L}_{GD}[\phi] + \frac{\alpha}{4B} \sum_{b=1}^B \left\| \frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi} \right\|^2 \\ &= L[\phi] + \frac{\alpha}{4} \left\| \frac{\partial L}{\partial \phi} \right\|^2 + \frac{\alpha}{4B} \sum_{b=1}^B \left\| \frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi} \right\|^2 \end{aligned}$$

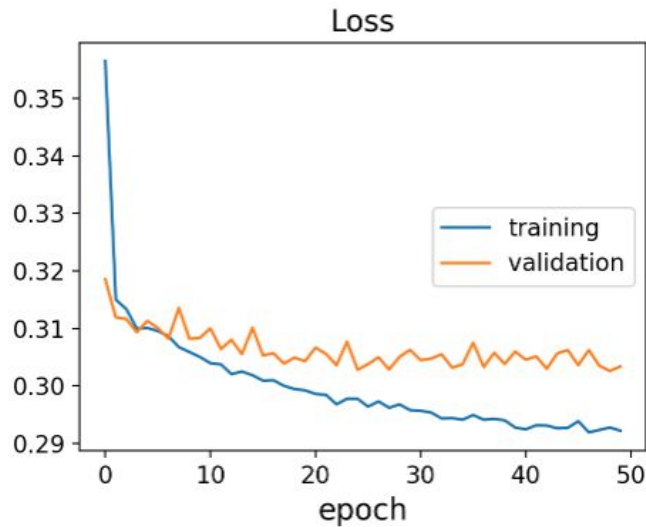
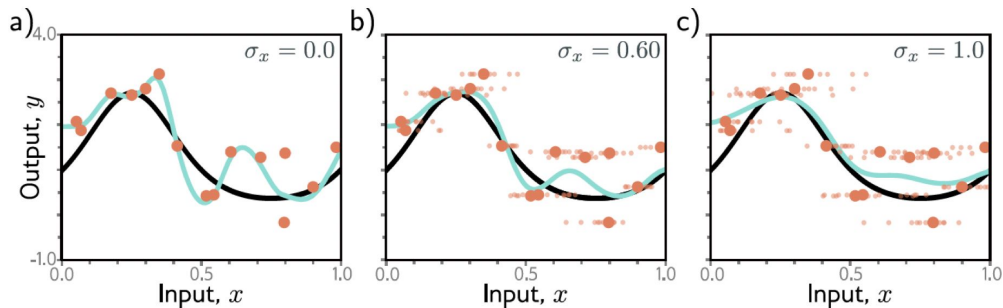


Other strategies:

Early stopping: Stop training before overtraining happens!

Ensembling: Take the average of multiple independent models

Adding noise / augmentations: Smooths out your function



Peer review

How to give a good peer review

- Don't expect to simply read it through once carefully and then give good feedback
- Define your role: to help improve this paper, or to filter the best papers through?
- Always follow specific guidelines (if you've been given them)
- After your first 1-2 passes, **start by summarizing the paper in your own words**
 - The authors should find your summary largely uncontroversial.
- Imagine your friend wrote it: look for the strengths first
- Be kind, professional, and honest.
- Consider using first-person: "I found Section 3.4 to be unclear..."
- Make your comments **specific and actionable** (but keep a big-picture perspective, too)
- Do the results match up to the claims of the paper?
- Are there missing references or areas that will strengthen the paper?
- **Organize your thoughts:**
 - Summary, Strengths, Weaknesses, Questions for the Authors
 - Summary, Major comments, Minor comments

In-class work

Peer review of a paper!

We'll come back 10 minutes before the end of class to discuss it.